# Requirement Analysis Technology Stack
# (Architecture & Stack)

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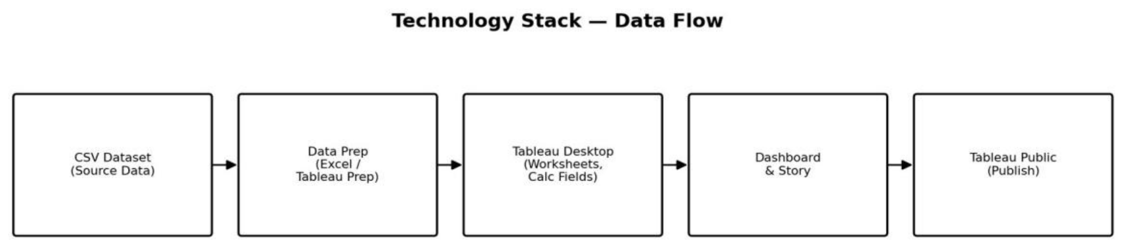
|---|---|
| **Date** | 30 Jun 2026 |
| **Team ID** | |
| **Project Name** | Food Ordering Behaviour & Customer Trends: A Structured Analysis of Choices and Habits |
| **Maximum Marks** | 4 Marks |

## Technology Stack Architecture Diagram




## Table-1: Components & Technologies

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | **Data Source / User Interface** | Interactive Tableau Dashboard & Story in a web browser | Tableau Dashboard, Tableau Story, Tableau Public (Web) |
| 2. | **Data Extraction Logic** | Reading and validating the raw order dataset before use | Python / Pandas (for pre-analysis), CSV |
| 3. | **Data Preparation Logic** | Cleaning, type casting and creating derived fields (age group, revenue, repeat-order flag) | Tableau Data Prep / Excel |
| 4. | **Visualization Logic** | Building calculated fields, worksheets and aggregations | Tableau Desktop (Calculated Fields, LOD Expressions) |
| 5. | **Database** | Data type and configuration of the underlying dataset | Flat-file CSV (50,000 rows × 19 columns) |
| 6. | **Cloud Database** | Extract storage used for fast querying | Tableau Hyper Extract (.hyper) |
| 7. | **File Storage** | Workbook & data source storage | Packaged Tableau Workbook (.twbx), Local Filesystem |
| 8. | **External API** | Not applicable  self-contained dataset | N/A |
| 9. | **Machine Learning Model** | Not used in this phase  insights via aggregation & visual analytics | N/A (Future Scope) |
| 10. | **Infrastructure** | Built locally on Tableau Desktop; published to cloud for public access | Local Desktop + Tableau Public Cloud |
| 11. | **Open-Source Frameworks** | Pandas/Python for initial data exploration; core visualization built natively in Tableau | Python, Pandas (pre-processing only) |

## Table-2: Application Characteristics

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | **Open-Source Frameworks** | Pandas/Python used for initial data exploration; core visualization built natively in Tableau | Python, Pandas |
| 2. | **Security Implementations** | Published as read-only public view; only anonymised user_id exposed; no PII in the dataset | Read-only Tableau Public |
| 3. | **Scalable Architecture** | Analytics pipeline architecture: CSV → Tableau Prep → Hyper Extract → Dashboard → Tableau Public. Scales by adding records/cities to CSV | Tableau Hyper Extract |
| 4. | **Availability** | Hosted on Tableau Public — Salesforce CDN — 99.9% uptime, accessible from any browser worldwide without installation | Tableau Public CDN |
| 5. | **Performance** | Tableau Hyper engine renders 50,000 records in under 5 seconds. Filters update charts in under 2 seconds on standard hardware | Tableau Hyper Engine |